

Customer & Revenue Performance Analysis of a UK Online Retailer

Business performance analysis of customer behaviour, revenue concentration and repeat purchasing for a UK-based online gift wholesaler, with data-driven recommendations for customer retention and revenue growth.

Dataset	UCI "Online Retail" dataset - 541,909 transactions, 2010-12-01 to 2011-12-09
Scope	Customer, revenue and business performance analysis (Intermediate Level)
Tools	Python (pandas, matplotlib, seaborn), reportlab

Executive Summary

This report analyses 522,568 cleaned sales lines from a UK-based online wholesaler of gift-ware, covering 2010-12-01 to 2011-12-09 (541,909 raw transaction lines before cleaning). Net revenue over the period was **£9.77M** from **19,773 orders** and **4,334 identified customers**, at an average order value of **£518** - a wholesale-sized basket, consistent with the company's many trade and gift-shop customers.

£9.77M Net revenue (gross – returns)	19,773 Orders	£518 Average order value	65% Customers who buy again
50% Revenue from Champions (14% of customers)	60% Revenue from top 10% of customers	38% Revenue earned in Sep-Nov alone	1.8% Genuine return rate

Five findings shape the recommendations in this report:

- **Revenue is highly seasonal.** September-November produced 38% of full-year net revenue; November alone was 2.3x the Jan-Aug monthly average, driven mainly by more orders and more customers, not bigger baskets.
- **Revenue is concentrated in a small core of customers.** The top 10% of customers generate 60% of net revenue, and “Champions” (601 customers, 14% of the base) alone generate 50%.
- **Most customers do come back.** 65% place more than one order and together drive 94% of revenue, but only 20% of new customers place a second order the following month.
- **Reported returns overstate the real return rate.** 62% of “returns” value comes from two orders cancelled minutes after being placed. Once these order-entry reversals are removed, the genuine return rate is 1.8%, not the headline 4.6%.
- **International revenue looks broader than it is.** The UK is 84.8% of net revenue; three of the next five markets by revenue (Netherlands, EIRE, Australia) each depend on one or two large accounts rather than a broad customer base.

1. Data & Methodology

Dataset. The UCI Online Retail dataset (Chen, Sain & Guo, 2012), covering all transactions of a UK-registered, non-store online retailer of unique all-occasion gifts between 2010-12-01 and 2011-12-09. Many of its customers are wholesalers, which explains the large basket sizes seen throughout this report.

Cleaning approach. Of 541,909 raw transaction lines: 5,268 exact duplicates were removed; 2,910 non-merchandise lines (postage, carriage, manual adjustments, bank charges, Amazon fees, gift vouchers) were set aside; 8,668 cancelled-order lines were separated into a returns table; and roughly 2,500 lines with a non-positive quantity or price were removed as stock write-offs rather than sales. This left 522,568 clean sales lines. The full, row-by-row log is in *outputs/tables/data_cleaning_log.csv* and *docs/methodology.md*.

Handling missing customer IDs. 25.1% of sales lines (14.7% of revenue) have no CustomerID - these are guest-style orders. They are kept in all revenue and product analysis, and excluded only from customer-level analysis (RFM, cohorts, repeat behaviour), which is based on the 4,334 identified customers covering 85.3% of revenue.

Order-entry reversals. A return is flagged as a same-day reversal when the same customer returns the exact product and quantity they bought within the previous 24 hours. These behave like corrections rather than genuine dissatisfaction, and account for 62% of all return value (driven mostly by two large cancelled orders). They are shown separately throughout, never silently merged into “returns”.

Key definitions.

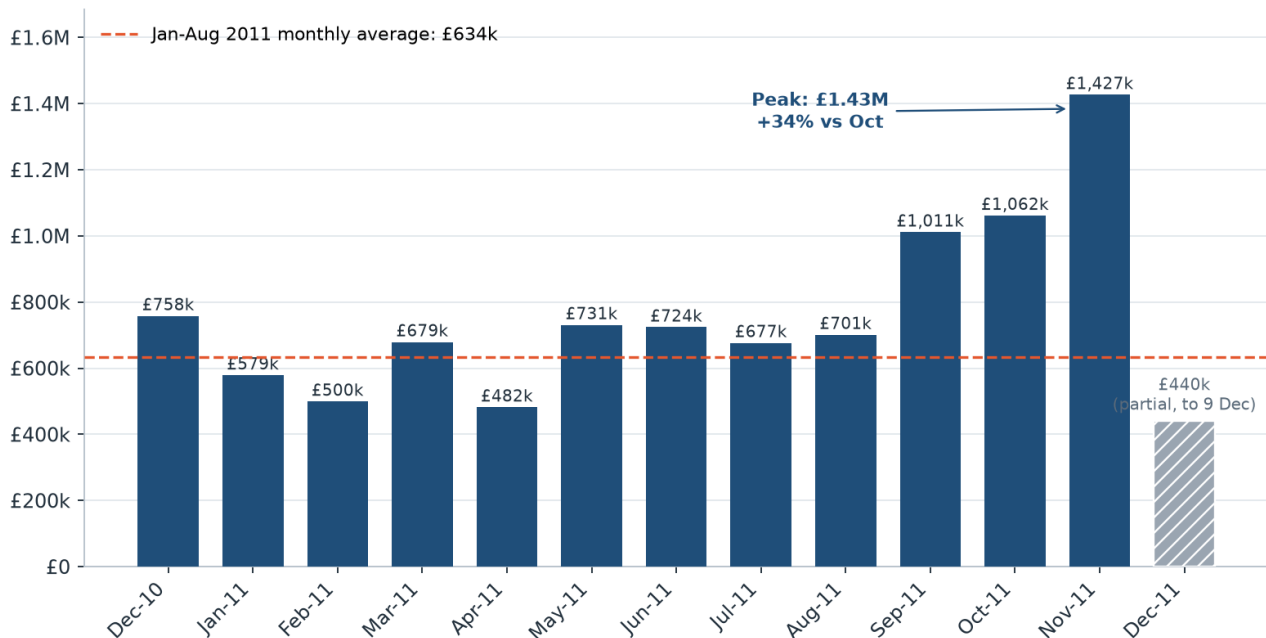
- **Gross revenue** = Quantity × Unit Price, summed over valid sales lines.
- **Net revenue** = Gross revenue – Returns value (the headline revenue metric used throughout).
- **Order** = one invoice. **AOV** = gross revenue / number of orders.
- **RFM segment** = Recency / Frequency / Monetary scoring (1-5 each) computed for the 4,320 identified customers with positive net revenue; see Section 5 for the segment definitions.

2. Revenue Trend & Seasonality

Net revenue rose from a Jan-Aug 2011 monthly average of £634,153 to a peak of £1.43M in November 2011 - a rise of 2.3x. December 2011 data stops on the 9th and is shown hatched; it is a partial month and is excluded from all month-over-month comparisons.

Net revenue by month: a strong autumn peak

Sep-Nov generated 38% of full-year net revenue (25% would be flat). Dec-11 is incomplete and not comparable.

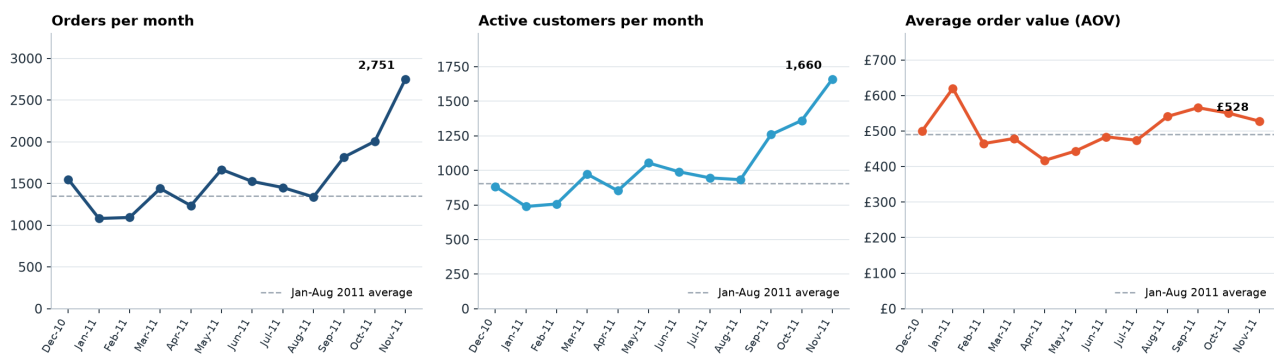


April 2011 was the weakest full month (£482,154), a 29% drop from March; May then recovered 52% month-on-month, the largest gain of the year.

What drove the November peak?

Breaking the peak into its components shows growth came overwhelmingly from **more orders and more customers**, not from customers spending more per order:

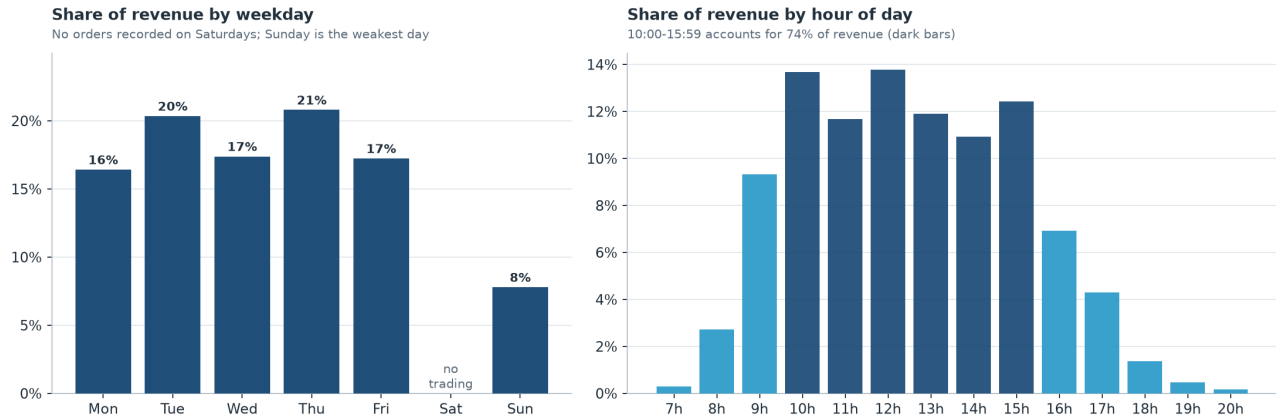
What drove the November peak? More orders (2.0x) from more customers (1.8x), not bigger baskets (AOV only +8%)



Orders in November were 2.0x the Jan-Aug average and active customers were 1.8x, while AOV rose only 8%. This points to **acquisition and reactivation** (new and returning shoppers placing orders) as the seasonal growth lever, rather than upsell or bigger baskets - a distinction that matters for how the autumn peak should be planned and staffed.

Trading patterns within the week

No orders are recorded on Saturdays in this dataset (a data or business-hours artefact, not necessarily a real trading gap), and Sunday is consistently the weakest trading day. Within the day, the 10:00-16:00 window concentrates the large majority of revenue, consistent with a business-hours, trade-buyer customer base.

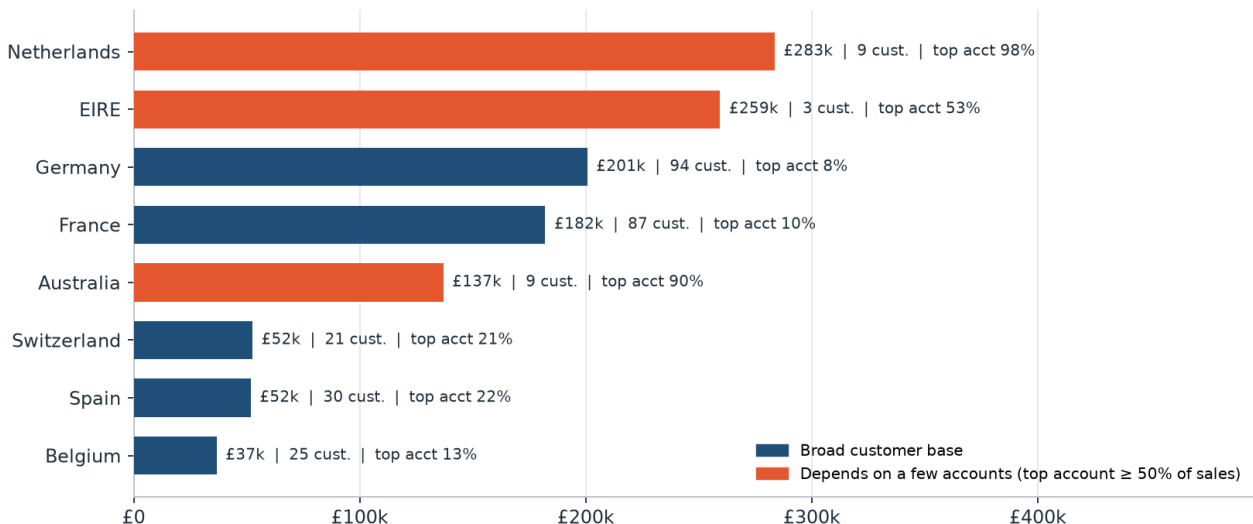


3. Geographic Performance

The UK accounts for 84.8% of net revenue across 3,916 customers. The remaining 15.2% is spread across 37 other countries, but revenue size is a misleading guide to market health: some of the largest “international markets” are really one or two large wholesale accounts.

International revenue: three of the top five markets hinge on a handful of accounts

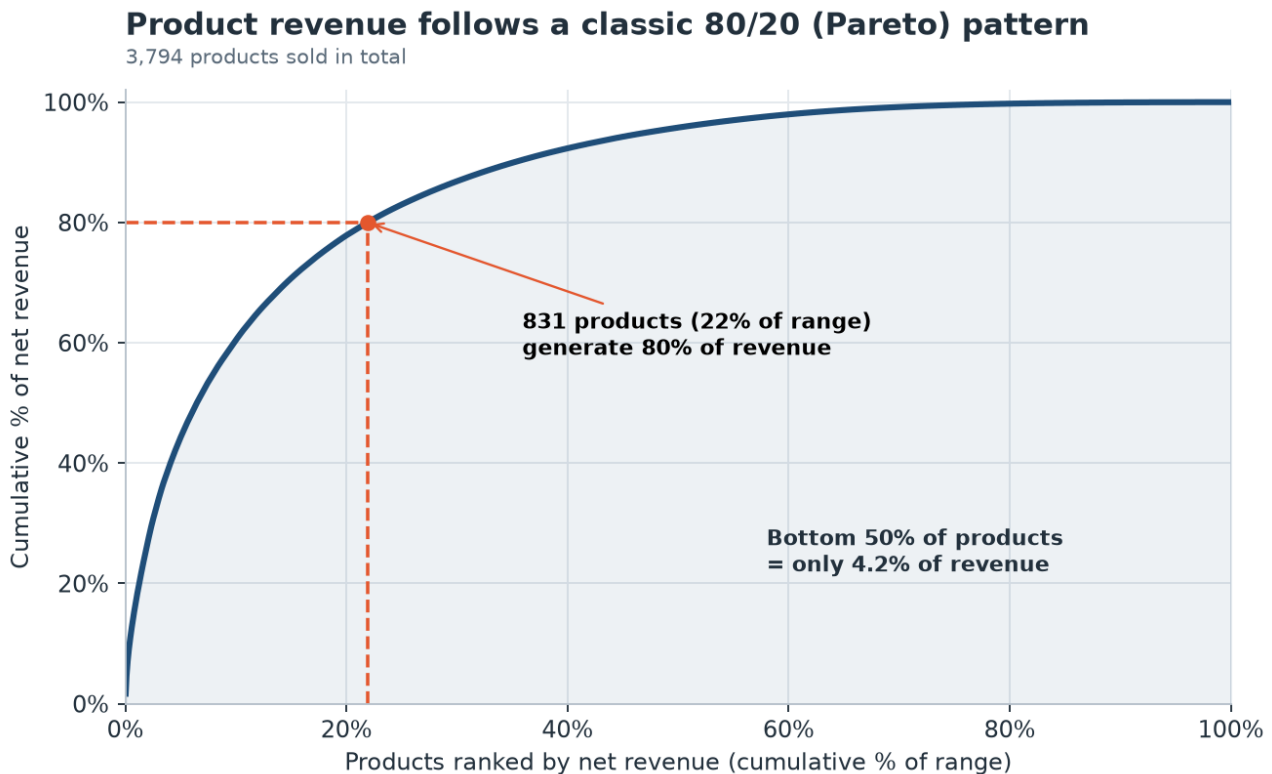
UK = 84.8% of net revenue. Netherlands, EIRE and Australia are each driven by 1-2 large accounts; Germany and France are broad-based.



Netherlands (98% from one account), EIRE (53% from one account) and Australia (90% from one account) are effectively key-account relationships dressed up as country performance - losing that single customer would collapse the market. By contrast, Germany and France have broad, diversified customer bases (257 combined customers across the five broad markets, at an average of £2,037 revenue per customer) and are the more reliable growth candidates.

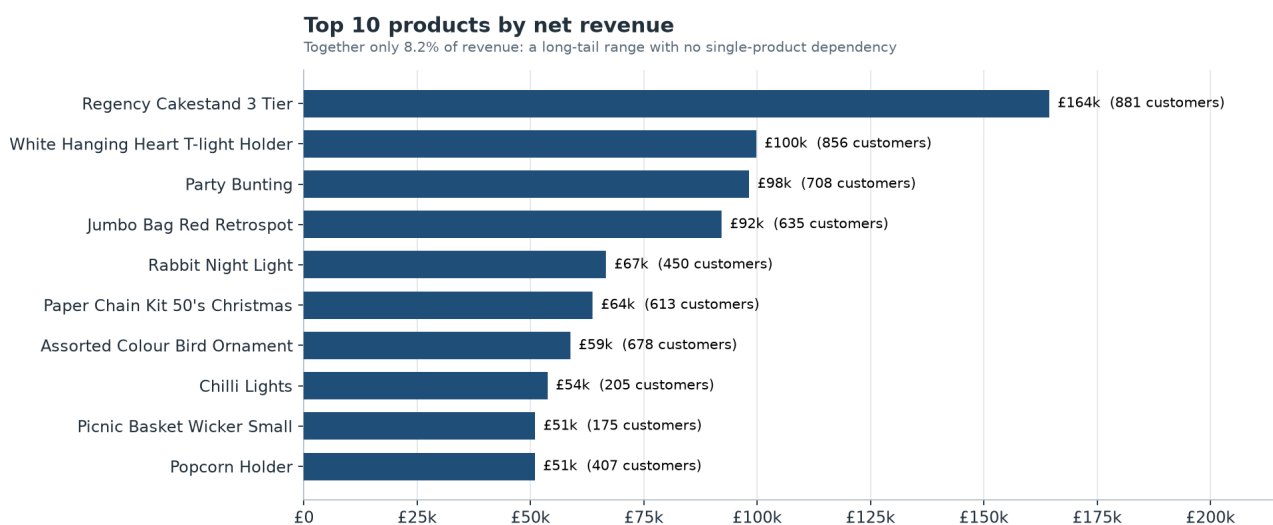
4. Product Performance

3,794 distinct products were sold. The top 10 products generate only 8.2% of net revenue - there is no single-product dependency - but the range as a whole follows a textbook Pareto pattern:



Just 831 products (22% of the range) generate 80% of revenue, while the bottom half of the range contributes only 4.2%. This is a candidate list for inventory rationalisation: the long tail carries working-capital and listing costs disproportionate to its revenue contribution.

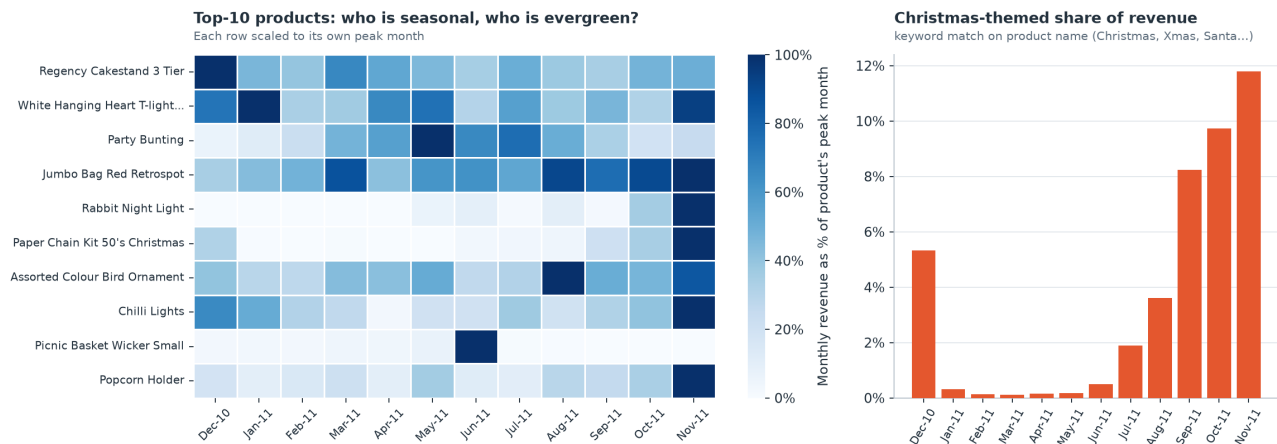
Top products by revenue



Which products are seasonal, and which are evergreen?

Scaling each of the top-10 products to its own peak month shows a mix of steady, year-round sellers (e.g. the cakestand and party bunting peak outside the holiday period) and clearly seasonal lines that spike in

November. A simple keyword screen for Christmas-themed products confirms the same pattern at the category level, rising from under 1% of monthly revenue in mid-year to nearly 12% in November.



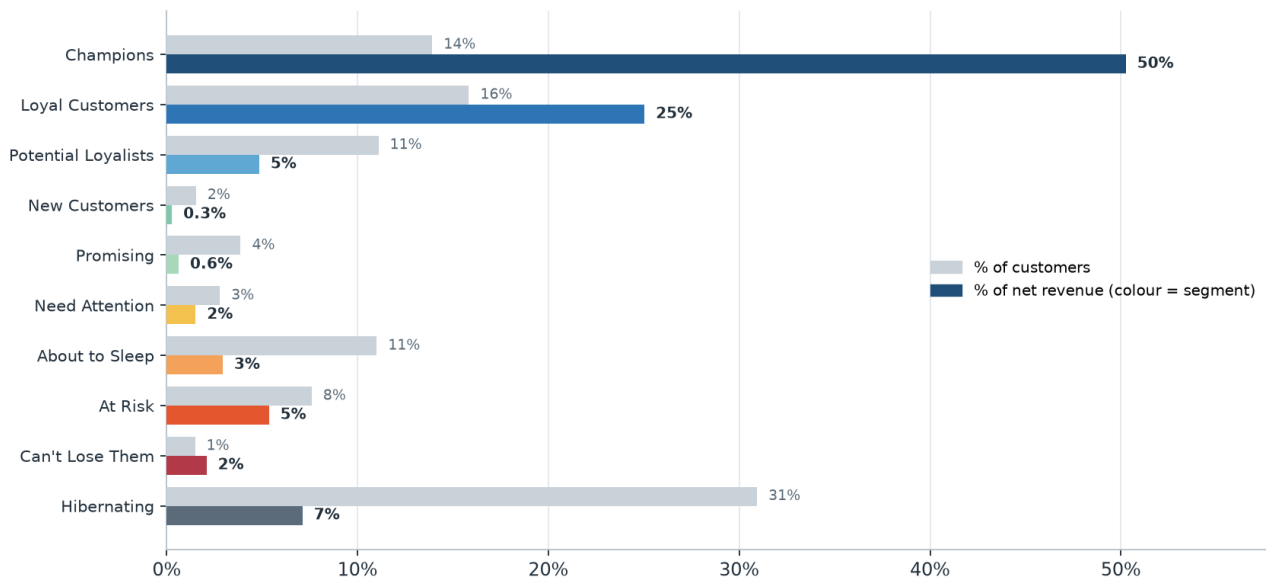
70 products ($\geq$ £5k annual revenue each, $\geq$ 60% of their revenue in Sep-Nov) are strongly seasonal, together worth £774k a year. These are the SKUs to prioritise for autumn stock-in and to de-prioritise (or discount) outside the season.

5. Customer Segmentation (RFM)

Every identified customer with positive net revenue was scored on Recency (days since last order), Frequency (number of orders, using business-defined bands) and Monetary value (net revenue), each on a 1-5 scale, and mapped to ten standard lifecycle segments.

RFM segments: customers vs. revenue

Champions are 14% of customers but 50% of revenue; At Risk / Can't Lose / Hibernating = 40% of customers, 15% of revenue

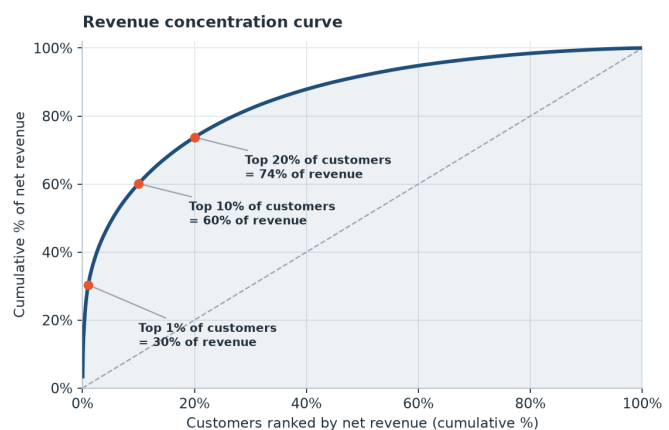
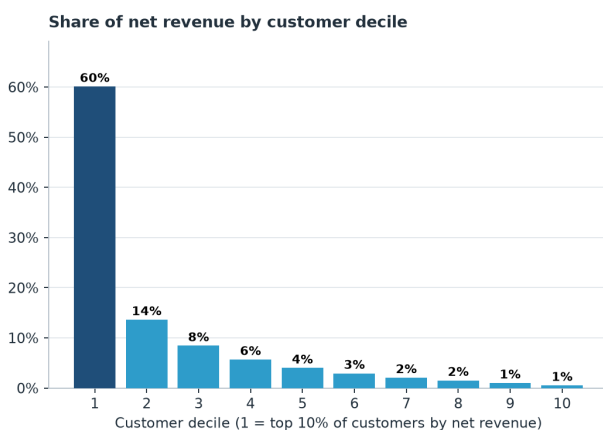


Champions (601 customers, 14% of the base) alone generate 50% of net revenue, and together with **Loyal Customers** (25% of revenue) these two segments carry the great majority of the business. At the other end, **At Risk**, **Can't Lose Them** and **Hibernating** together are 40% of customers but only 15% of revenue - representing £1.20M of historic revenue now at risk of being lost entirely, including 64 "Can't Lose Them" customers (previously high-value, now lapsing) worth £172k.

Revenue concentration

The RFM segments are one view of a deeper pattern: revenue is heavily concentrated in a small share of customers. The top 10% of customers generate 60% of net revenue, and just 234 customers (5% of the base) account for half of all revenue.

A small core of customers drives the business: top 10% = 60% of revenue, bottom 50% = 8%

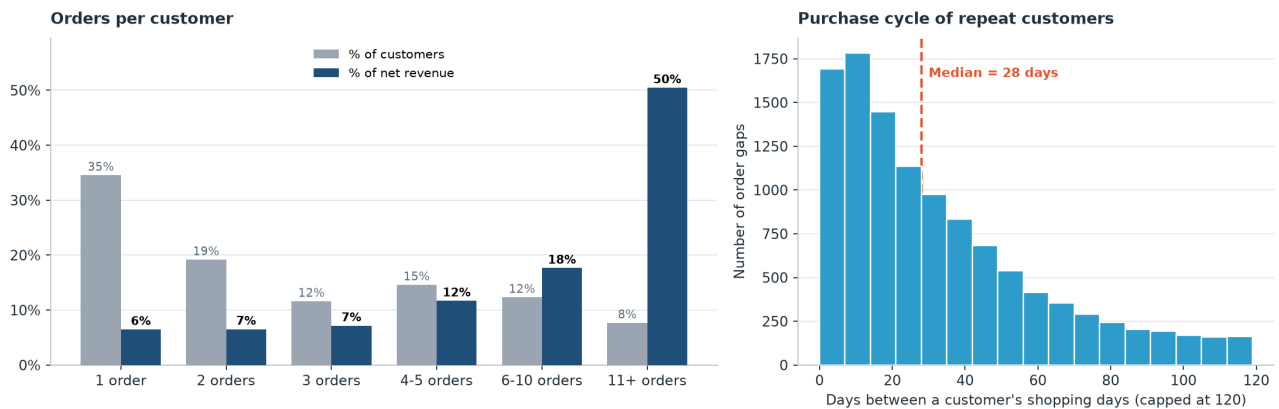


This concentration is healthy for focused account management, but it is also a concentration risk: the single largest customer alone represents 3.4% of identified net revenue.

6. Repeat Purchase Behaviour & Retention

65% of identified customers place more than one order, and these repeat customers generate 94% of net revenue - on average 7.6x more revenue per customer than one-time buyers. For those who do come back, the typical (median) gap between shopping days is 28 days.

65% of customers buy again, and they generate 94% of revenue



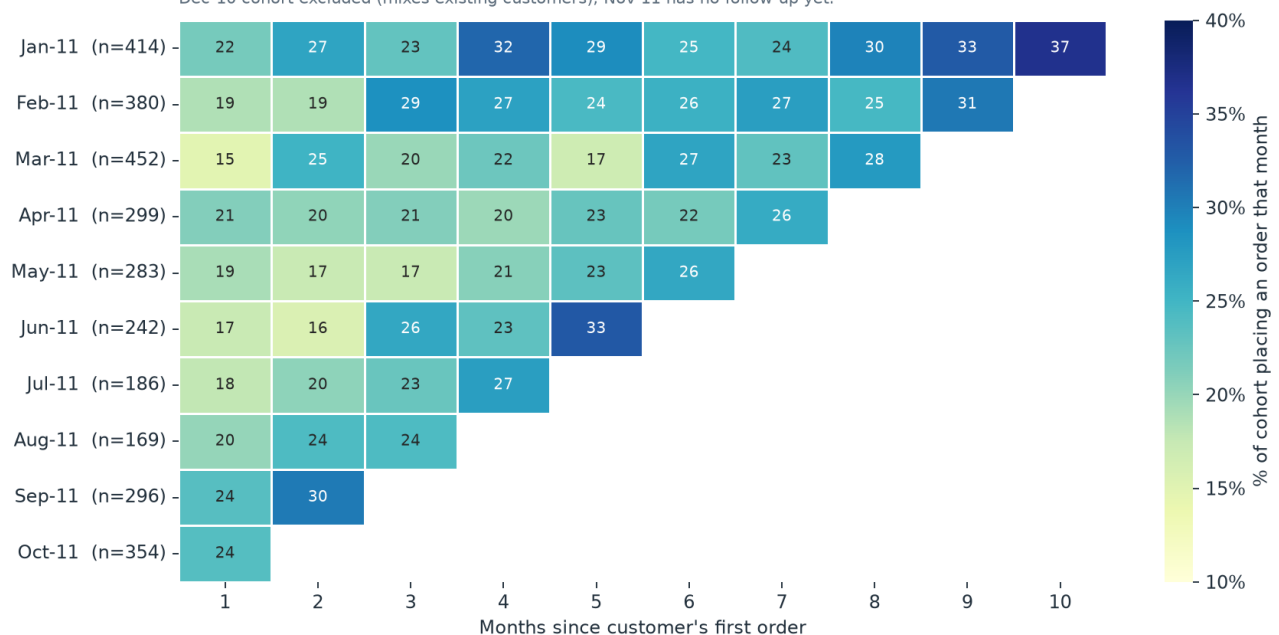
Revenue is not just concentrated by segment, but by order frequency: customers with 11 or more orders are only 8% of the base yet generate half of net revenue, while one-time buyers - over a third of customers - generate just 6%.

Do new customers come back? Cohort retention

Tracking each month's new-customer cohort forward, only 20% place another order the following month. Retention does not then decay steadily - it plateaus around 23-25% for months 2 through 7 - suggesting the biggest opportunity is not slowing an ongoing decline, but converting more first-time buyers into a second purchase in the critical first month.

Cohort retention: about 1 in 5 new customers returns the next month

Avg. month-1 retention 20%; months 2-7 hold at 23-25% (no steady decay); the lift after month 7 coincides with the Sep-Nov peak. Dec-10 cohort excluded (mixes existing customers); Nov-11 has no follow-up yet.

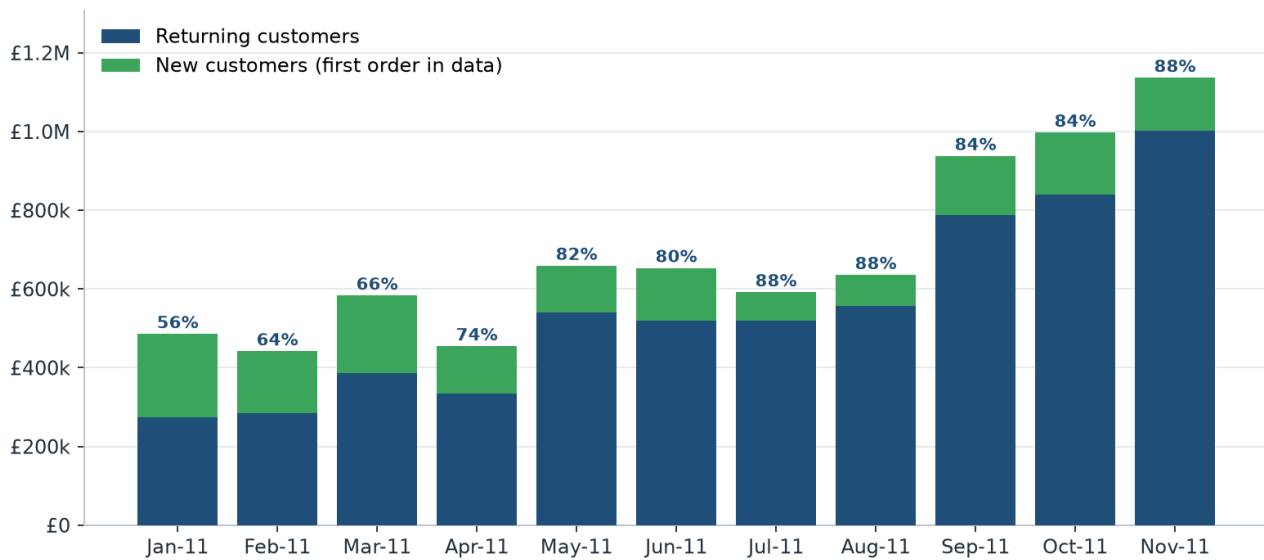


New vs. returning revenue

The share of monthly revenue coming from returning customers rose steadily across the year, from roughly half in January to 88% by November - additional evidence that the autumn peak was substantially fuelled by the existing, already-engaged customer base returning to buy again, alongside new customer acquisition.

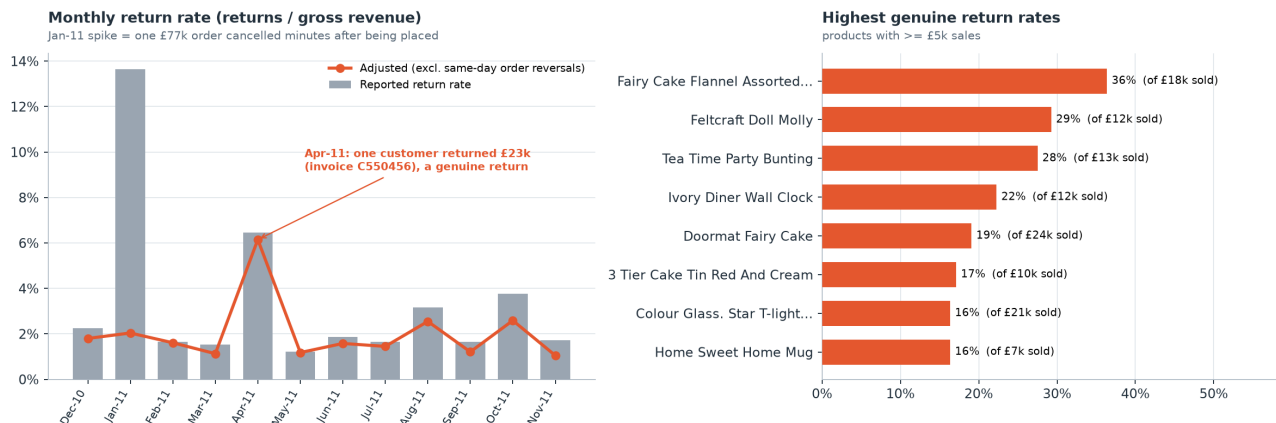
Gross sales from new vs. returning customers (customers with a customer ID)

Labels = returning share. Early-2011 'new' is overstated: the data starts Dec-10, so some 'new' customers are pre-existing.



7. Returns Analysis

The headline return rate (returns value / gross revenue) is 4.6%, but 62% of that value comes from just two orders that were cancelled within 24 hours of being placed by the same customer for the same product and quantity - almost certainly order-entry corrections rather than genuine returns. Excluding these, the **genuine return rate is 1.8%**, a materially healthier and more decision-useful figure.



The clearest genuine quality signal is the “Fairy Cake Flannel” line, with a 36% return rate on meaningful sales volume, followed by several other homeware and gift items in the 15-30% range - these are worth a quality or listing-description review.

8. Recommendations

The following five actions follow directly from the findings above, in priority order:

1. Protect and grow the Champions segment

601 customers generate 50% of revenue. Introduce a named account-management or loyalty-tier programme for this group, with proactive contact ahead of the autumn peak and early access to seasonal stock, to defend this concentrated revenue base.

2. Run a targeted win-back campaign for lapsing high-value customers

£1.20M of historic revenue sits with At Risk / Can't Lose / Hibernating customers. Prioritise the 64 "Can't Lose Them" accounts (previously high-value, now lapsing) with personal outreach and an incentive, before attempting the larger, lower-value Hibernating group.

3. Build a first-30-day reorder journey for new customers

Only 20% of new customers reorder the following month, and the typical repeat cycle is 28 days. A structured onboarding sequence (e.g. a second-order incentive timed to arrive around day 20-25) targets the single biggest drop-off point in the customer lifecycle.

4. Plan inventory, staffing and cash flow around the Sep-Nov peak

November revenue is 2.3x the Jan-Aug average, driven by order and customer volume rather than basket size. Begin stock build and seasonal staffing from September, prioritising the 70 identified seasonal SKUs, and use the Jan-Apr trough for clearance and demand-generation promotions.

5. Grow broad-based international markets, manage key-account risk in others

Germany and France have wide, diversified customer bases and room to grow with normal marketing investment. Netherlands, EIRE and Australia depend on one or two accounts each - treat these as key-account relationships requiring retention planning, not organic growth markets, and avoid over-reading their revenue size as market health.

Appendix: Supporting Detail Tables

Data Cleaning Summary

#	Cleaning rule	Rows affected	Rows remaining
0	Raw data loaded	0	541,909
1	Remove exact duplicate rows (all 8 columns identical)	5,268	536,641
2	Separate non-product lines (POST, DOT, M, D, S, B, C2, BANK CHARGES, A...	2,910	533,731
3	Separate cancellations (InvoiceNo starts with 'C') into returns table	8,668	525,063
4	Remove sales lines with Quantity <= 0 (stock write-offs / damages)	1,336	523,727
5	Remove sales lines with UnitPrice <= 0 (free items / adjustments)	1,159	522,568
6	Remove return lines with UnitPrice <= 0 or Quantity >= 0	0	8,668
7	Standardise product names (one canonical description per StockCode)	6,894	522,568
8	Add derived columns: Revenue, InvoiceMonth, InvoiceDay, Weekday, Hour,...	522,568	522,568
9	Flag lines without CustomerID (kept for revenue/product analysis)	131,418	522,568
10	FINAL clean sales table	0	522,568

RFM Segment Summary

Segment	Customer s	% cust.	Net revenue	% revenue	Avg recency (days)
Champions	601	13.9%	£4,155,828	50.2%	6
Loyal Customers	684	15.8%	£2,070,009	25.0%	33
Potential Loyalists	479	11.1%	£400,631	4.8%	17
New Customers	66	1.5%	£22,305	0.3%	8
Promising	167	3.9%	£50,930	0.6%	24
Need Attention	120	2.8%	£124,550	1.5%	50
About to Sleep	475	11.0%	£242,569	2.9%	53
At Risk	328	7.6%	£442,765	5.3%	135
Can't Lose Them	64	1.5%	£171,745	2.1%	132
Hibernating	1,336	30.9%	£588,305	7.1%	209

Top 10 Countries by Net Revenue

Country	Net revenue	% of total	Customers	AOV	Top account %
United Kingdom	£8,282,076	84.8%	3,916	£487	4%
Netherlands	£283,480	2.9%	9	£3,053	98%
EIRE	£259,380	2.6%	3	£960	53%
Germany	£200,620	2.0%	94	£464	8%
France	£182,077	1.9%	87	£483	10%
Australia	£136,922	1.4%	9	£2,466	90%
Switzerland	£52,483	0.5%	21	£1,061	21%
Spain	£51,747	0.5%	30	£633	22%
Belgium	£36,663	0.4%	25	£377	13%
Japan	£35,420	0.4%	8	£1,969	57%

Top 10 Products by Net Revenue

Product	Net revenue	Units	Orders	Return rate
Regency Cakestand 3 Tier	£164,459	13,851	1,988	5.6%
White Hanging Heart T-light Holder	£99,791	37,933	2,265	6.2%
Party Bunting	£98,244	18,283	1,685	1.2%
Jumbo Bag Red Retrospot	£92,176	48,371	2,089	2.1%
Rabbit Night Light	£66,662	30,739	994	0.3%
Paper Chain Kit 50's Christmas	£63,715	19,329	1,160	1.8%
Assorted Colour Bird Ornament	£58,792	36,362	1,455	0.2%
Chilli Lights	£53,747	10,302	661	0.7%
Picnic Basket Wicker Small	£51,024	1,937	463	0.8%
Popcorn Holder	£50,968	56,898	1,392	0.7%

Full detail tables (22 CSVs covering every metric in this report) are provided in *outputs/tables/*; full-resolution charts are in *outputs/charts/*; the one-page executive dashboard is in *dashboard/executive_dashboard.png*; and the complete, reproducible analysis code is in *src/*. See *README.md* for how to reproduce every figure in this report from the raw data.